

COL DU GRAND ST BERNARD, Switzerland

WMO: 067170

Lat: 45.8689N

Lon: 7.1708E

Elev: 8133

StdP: 10.86

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-0.6	2.6	-19.2	2.6	15.4	-15.7	3.2	15.4	41.2	18.3	36.4	17.1	15.9	90	0.900

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	62.0	49.0	59.5	48.2	57.1	47.4	51.7	57.5	50.3	56.1	49.1	54.9	10.4	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
49.5	71.2	53.3	48.0	67.2	52.3	46.4	63.3	51.3	24.4	57.4	23.4	56.2	22.6	55.0	55.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
34.6	29.3	25.4	DB	-4.0	65.6	4.2	2.5	-7.0	67.4	-9.5	68.8	-11.8	70.2	-14.8	72.1	
			WB	-4.7	54.0	3.9	0.8	-7.5	54.6	-9.8	55.0	-11.9	55.5	-14.7	56.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	32.1	18.9	17.9	22.4	28.1	34.4	44.2	47.7	47.4	40.9	35.3	25.9	21.3
	DBStd	12.85	7.74	8.45	7.54	6.37	6.62	7.26	6.38	6.65	6.60	7.37	7.90	7.68
	HDD50	6651	964	900	856	656	484	202	117	128	277	455	723	889
	HDD65	12001	1429	1320	1321	1106	948	625	537	546	722	920	1173	1354
	CDD50	123	0	0	0	0	0	27	45	47	4	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
	WSAvg	11.9	12.4	12.6	12.5	12.4	12.2	10.8	10.8	10.5	11.4	11.9	12.8	12.3
	PrecAvg	47.6	4.2	3.9	3.4	3.8	4.4	3.9	4.0	3.7	3.2	3.7	4.9	4.6
(15) Precipitation	PrecMax	65.2	10.4	11.1	10.5	6.3	8.9	7.8	7.3	7.2	7.6	9.6	15.1	14.8
	PrecMin	35.8	0.7	0.5	0.5	0.7	1.4	1.1	1.7	1.3	0.7	0.7	1.3	0.0
	PrecStd	6.5	2.9	2.9	2.4	1.7	2.2	1.6	1.4	1.5	1.6	2.1	3.2	3.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.2	39.9	40.0	44.6	53.9	64.7	65.5	65.0	57.6	52.5	46.5	38.2
		MCWB	28.0	27.9	30.0	35.1	43.5	48.2	49.7	49.1	46.0	41.0	33.7	27.6
	2%	DB	34.8	36.0	37.0	41.8	49.8	60.8	61.6	61.9	54.6	49.0	41.4	35.1
		MCWB	26.5	26.1	28.2	34.0	41.9	47.8	49.0	49.6	45.0	39.1	32.3	26.5
	5%	DB	32.4	32.4	34.4	39.4	46.5	57.7	59.0	59.1	52.1	46.7	38.7	33.1
		MCWB	25.9	25.1	27.5	33.3	40.0	47.0	48.2	48.4	44.1	38.0	31.2	25.8
	10%	DB	29.9	29.1	32.2	36.9	43.6	54.7	56.6	56.6	49.6	44.3	35.9	31.2
		MCWB	25.0	24.1	26.8	32.0	38.2	46.4	47.7	47.5	42.8	36.7	29.8	25.2
	0.4%	WB	31.9	30.4	32.6	37.5	45.0	52.0	53.7	53.2	49.2	43.8	36.9	31.9
		MCDB	33.4	34.0	35.8	40.7	49.9	58.0	58.7	58.8	53.7	48.0	40.5	33.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	29.6	28.4	30.6	35.4	42.9	50.2	51.8	51.8	47.3	42.0	34.7	29.9
		MCDB	31.9	32.8	33.6	39.5	48.5	56.2	57.2	57.9	51.7	45.7	39.0	32.3
	5%	WB	27.3	26.4	29.2	34.1	40.9	48.9	50.3	50.3	45.8	40.6	32.5	28.1
		MCDB	30.7	31.8	32.3	38.4	45.7	55.2	56.1	55.9	50.0	44.0	36.4	30.5
	10%	WB	25.0	24.1	27.9	32.5	38.9	47.1	49.1	48.9	44.2	39.1	31.1	26.3
		MCDB	29.3	28.8	31.0	36.3	42.5	53.6	54.9	54.6	48.4	42.5	34.2	29.2
(35) Mean Daily Temperature Range	5% DB	MDBR	8.0	8.1	8.3	8.1	8.6	10.3	11.1	10.4	9.3	8.2	7.4	8.1
		MCDBR	9.7	10.2	9.7	9.8	11.4	12.8	13.2	12.2	11.4	10.4	8.7	9.0
		MCWBR	7.7	7.2	6.7	6.4	6.4	7.0	7.2	7.2	7.0	7.3	5.7	6.4
	5% WB	MCDBR	9.6	10.2	8.4	8.7	10.6	12.0	12.1	11.3	10.2	9.0	7.7	8.7
		MCWBR	9.1	8.3	6.8	6.2	6.4	7.0	7.4	6.9	6.5	7.1	6.0	8.1
(40) Clear-Sky Solar Irradiance	taub	0.203	0.220	0.252	0.284	0.297	0.302	0.298	0.290	0.276	0.259	0.229	0.204	
	taud	2.396	2.359	2.335	2.386	2.444	2.464	2.470	2.505	2.539	2.572	2.533	2.439	
	Ebn at Noon	302	313	314	311	308	305	305	305	302	295	289	291	
	Edh at Noon	22	28	34	35	35	34	34	31	27	23	19	18	
	RadAvg	491	766	1112	1396	1597	1904	1923	1676	1304	846	496	408	
(45) All-Sky Solar Radiation	RadStd	56	99	146	178	135	157	149	122	104	95	72	56	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (87 neighbors)	N/A	N/A	N/A	+1.22	+0.67	+0.47	N/A	N/A	+101	+32

Nomenclature: See separate page